

## 2026 Evil Swerve Drive Todo

| File             | Todo                                                                                                                                                                                                                     | Priority | Priority Numb | Status  |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------|---------|
| IntakeSystem     | determine up and down angles on the robot -- suggest starting with 90 is up and 0 is down...                                                                                                                             | Low      | 4             | Partial |
| SwerveModule     | The NEO Vortex is NOT use spark max. I think it uses a Spark Flex controller. These are different. Need to read and convert.                                                                                             | High     | 1             | Open    |
| TurretLauncher   | Add speed control for launcher. One PID will create output value for both motors. Add motors, Add sparkmax encoders, create average velocity, add simple motor feedforward and PID. (see swervemodule drive controller.) | High     | 1             | Open    |
| TurretLauncher   | Convert to degrees. Think that the default is rotations. Also take into account gear ratio...                                                                                                                            | High     | 1             | Open    |
| AutoSystem       | Add code to the init function to get all available autos, then call the function to read them into the auto cache...                                                                                                     | High     | 2             | Open    |
| IntakeSystem     | scale arm position.. default is rotations.... so $DEG = ROT * 360 / GEAR\_RATIO$                                                                                                                                         | High     | 2             | Open    |
| TrajectorySystem | Make this modulo +/- PI... There is a WPILIB function for this... MathUtil.angleModulus                                                                                                                                  | High     | 2             | Open    |
| AgitatorSystem   | Set motor CAN Id                                                                                                                                                                                                         | Medium   | 3             | Open    |
| AutoSystem       | add locNTSsend trigger update.                                                                                                                                                                                           | Medium   | 3             | Open    |
| AutoSystem       | Add items to NT tables for debugging and watching autos. Suggest Step, Timeout, Parm1, Parm2, Parm3, Function. Add public getters for these.                                                                             | Medium   | 3             | Open    |
| FeederSystem     | Set feeder motor CAN ID                                                                                                                                                                                                  | Medium   | 3             | Open    |
| SwerveTeleop     | Add a driver button to execute a trajectory while button is held done. When first pressed (edge triggered) set init variable in call to follow trajectory.                                                               | Medium   | 3             | Open    |
| SwerveVision     | will have to set physical camera offset position once it is known. There is a spreadsheet in programming slack channel.                                                                                                  | Medium   | 3             | Open    |
| TrajectorySystem | add network table entries for ontarget, done, havesample, timelengthoftrajectory (rename if you don't like name)                                                                                                         | Medium   | 3             | Open    |
| TrajectorySystem | TODO: add getters for new things written to network tables. ontarget, havesample, done.                                                                                                                                  | Medium   | 3             | Open    |
| TurretLauncher   | TODO: Set CAN ID                                                                                                                                                                                                         | Medium   | 3             | Open    |
| AgitatorSystem   | Remove unused imports                                                                                                                                                                                                    | Low      | 4             | Open    |
| AutoRoutine      | Remove unused imports                                                                                                                                                                                                    | Low      | 4             | Open    |
| Climb            | Do the TODO... We don't know what climb is yet.                                                                                                                                                                          | Low      | 4             | Open    |

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|------------------|-------------------------------------------------------------------------------------------------------------------------------|----------|---|----------|
| FeederSystem     | Remove unused imports.                                                                                                        | Low      | 4 | Open     |
| FeederSystem     | Might need to do speed control here...                                                                                        | Low      | 4 | Open     |
| IntakeSystem     | Say which motor is arm and which is intake.... Is 1 intake, and 2 arm ??                                                      | Low      | 4 | Open     |
| IntakeSystem     | config values need to be changed/tuned                                                                                        | Low      | 4 | Open     |
| IntakeSystem     | This appears to be redundant.                                                                                                 | Low      | 4 | Open     |
| IntakeSystem     | Is this intake or arm motor ????                                                                                              | Low      | 4 | Open     |
| IntakeSystem     | what is this for? Maybe check the actual angle -- encoderRot -- instead.                                                      | Low      | 4 | Open     |
| SwerveDrive      | Remove unused imports                                                                                                         | Low      | 4 | Open     |
| SwerveModule     | Remove unused imports                                                                                                         | Low      | 4 | Open     |
| SwerveOdometry   | put some comments here. For example, this is the initial robot position (pose)                                                | Low      | 4 | Open     |
| SwerveTeleop     | Remove old commented out things that will never be used.                                                                      | Low      | 4 | Open     |
| SwerveTeleop     | Add driver button box ???                                                                                                     | Low      | 4 | Open     |
| SwerveTeleop     | Add aux button box ???                                                                                                        | Low      | 4 | Open     |
| TrajectorySystem | Remove unused imports.                                                                                                        | Low      | 4 | Open     |
| TurretLauncher   | Remove unused imports                                                                                                         | Low      | 4 | Open     |
| AgitatorSystem   | Add JavaDoc                                                                                                                   | Very Low | 5 | Open     |
| AutoRoutine      | Add JavaDoc                                                                                                                   | Very Low | 5 | Open     |
| AutoStep         | Suggest adding the following Cmd -- ExecuteCmd (command name is in Function string variable)                                  | Very Low | 5 | Open     |
| AutoStep         | Suggest adding the following Cmd -- ExecuteCheckFor (command name is in function string variable )                            | Very Low | 5 | Open     |
| AutoStep         | Suggest adding the following Cmd -- AutoDone -- This will do nothing forever....similar to wait but with a forever timeout... | Very Low | 5 | Open     |
| AutoStep         | Add JavaDoc                                                                                                                   | Very Low | 5 | Open     |
| AutoSystem       | Add doc -- call this just prior to running ANY auto... Robot.java has a place for this. It isn't the overall just once init.  | Very Low | 5 | Open     |
| AutoSystem       | Does not compile. Replace with generic Exception instead of FileNotFoundException.                                            | High     | 1 | Complete |
| AutoRoutine      | set class variable with value of autoDescription parameter..                                                                  | High     | 2 | Complete |
| AutoRoutine      | set class variable with value of autoDescription parameter.. (2 <sup>nd</sup> constructor)                                    | High     | 2 | Complete |
| AutoSystem       | Add private static HashMap to store all trajectories <String,Trajectory<SwerveSample>                                         | High     | 2 | Complete |
| AutoSystem       | Add your newRoutine to the arrayList routines...                                                                              | High     | 2 | Complete |
| AutoSystem       | Add your newRoutine to the class variable to be created containing the HashMap                                                | High     | 2 | Complete |

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|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---|----------|
| AutoSystem       | If you store the routines in the class variable containing the HashMap, then this routine doesn't need to return anything (void)                                           | High   | 2 | Complete |
| AutoSystem       | Add a function to return an AutoRoutine from the hashmap with a calling String parameter of the name (file name without extension)                                         | High   | 2 | Complete |
| AutoSystem       | Add call to TrajectorySystem.FollowTrajectory                                                                                                                              | High   | 2 | Complete |
| IntakeSystem     | Use built in spark max encoder. We are no longer getting a through bore encoder..                                                                                          | High   | 2 | Complete |
| IntakeSystem     | Encoder definition isnt correct.. Now use spark max built in encoder .GetEncoder(); see above.                                                                             | High   | 2 | Complete |
| IntakeSystem     | limit switch is new wired to roboRIO... so this won't work. Use digital input.. Need to define object (class level), create object( in init function), read object (here). | High   | 2 | Complete |
| TrajectorySystem | add calculate of "complete" and return "complete" to user.<br>complete = elapsedtime >= trajectory time length AND onTarget.                                               | High   | 2 | Complete |
| TurretLauncher   | get the values of the goal position based on alliance. I think there is a spreadsheet in slack. For now just use RED target.                                               | High   | 2 | Complete |
| TurretLauncher   | This will not be throughbore encoder any longer. Use built in SparkMax encoder....                                                                                         | High   | 2 | Complete |
| TurretLauncher   | Use spark max built in encoder instead. Build didn't mount through bore encoder....                                                                                        | High   | 2 | Complete |
| TurretLauncher   | getPosition can be used above....                                                                                                                                          | High   | 2 | Complete |
| AutoFunctions    | Add new java file for static Auto processing functions.                                                                                                                    | Medium | 3 | Complete |
| AutoFunctions    | Add AutoFunctions.java file for static functions. Add an boolean AutoWait( ) function. It should set drive demand to zero and return false..                               | Medium | 3 | Complete |
| AutoSystem       | Add call to AutoFunctions.AutoWait().                                                                                                                                      | Medium | 3 | Complete |

[illegible]

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| Needed to add this to the individual case.... |
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